

From Scratch to Scalable: Building Smarter AI Agents with Frameworks



Jettro Coenradie



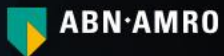
Daniël Spee



PLATINUM SPONSOR



GOLD SPONSORS



Rijksverheid

luminis. / part of yuma

X #jfall.



Jetro Coenradie

 /in/jetro



Daniël Spee

 /in/dspee

What are you going to do?

Create an Agent using Java + Spring Boot.

The prompt, ReAct, tools, memory, and multi-agents

Create an Agent using Spring AI.

Tools, memory, Guardrails, and multi-agents

Work with Evaluations.

LLM Evals, Human Evals and Accuracy

What are you going to do? (OPTIONAL)

Create an Agent with Embabel.

Tools, MCP

Secure your MCP Server.

OAuth

Schedule

14:40 - 15:00: Introduction + What are agents?

15:00 - 15:45: [Step_1] Work on Java Agent (Multi-agent is optional)

15:45 - 16:00: Multi-agent discussion + Spring AI Introduction

16:00 - 16:45: [Step_2] Work on Spring AI Agent (Multi-agent is optional)

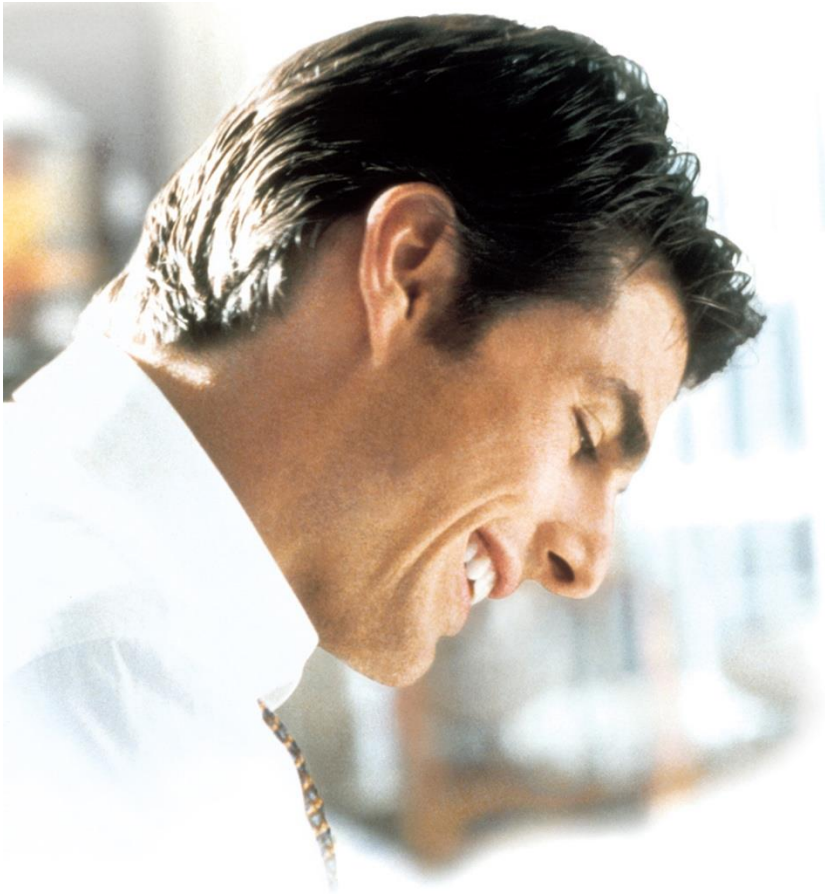
16:45 - 17:00: Evaluations + Guardrails

17:00 - 17:30: [Step_4, Step_5] Work on Evaluations and Guardrails

Optional: [Step_3, Step_6, Step_7] Work on MCP, Embabel Agent and secure MCP with Oauth

17:30 - 17:45: Wrap-up

What is an Agent?



Jerry Maguire ~ Sports Agent

One who is authorized to act for or in the place of another

~ Merriam Webster

A software agent is a program that acts for a user or another program.

~ Wikipedia

<https://www.merriam-webster.com/dictionary/agent>

https://en.wikipedia.org/wiki/Software_agent

What about AI Agents?

AI agents are software systems that use AI to pursue goals and complete tasks on behalf of users. They show reasoning, planning, and memory and have a level of autonomy to make decisions, learn, and adapt.

~ Google cloud

Key aspects of an Agent

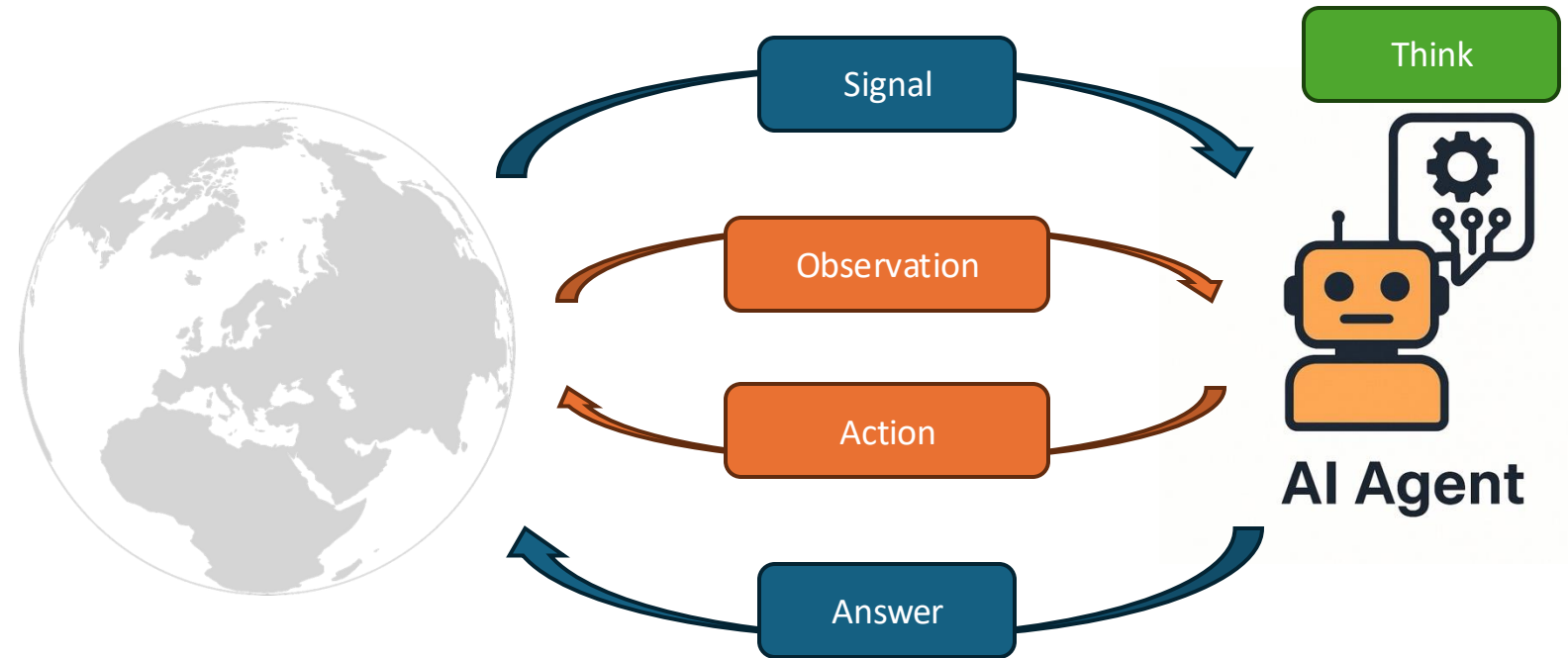
Reasoning

Acting

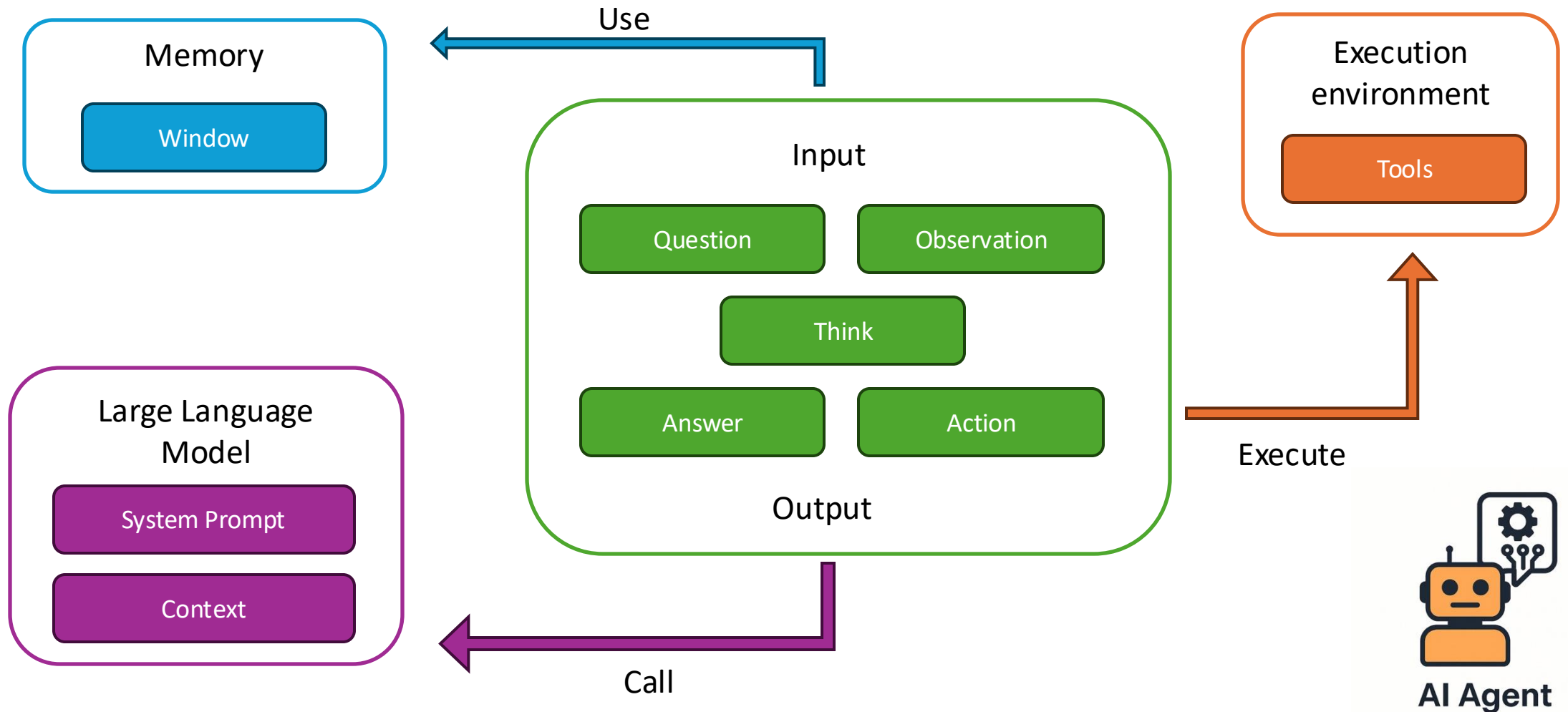
Observing

Planning

Collaborating



The Plain Java Agent



The assignments

- Check out the repository.
- Review the README.md file to become familiar with the project.
- The assignments are in the `Assignments` folder.
- Text blocks help you get familiar with the assignment.
- The code blocks tell you what to do.

<https://github.com/RAG4J/jfall-2025-assignments>

Project layout

Assignments: assignments markdown files

Data: data files for favourites and evals

Logs: logs for the favourites mcp server

Core-agent: files shared between all agents

Web-app: runs the UI for all agents.

Java-agent: implementation for Java agent

Springai-agent: implementation for Spring AI

Embabel-agent: implementation for Embabel

Favourites-mcp: MCP server

Eval-web-app: runs the evaluation web app

Connecting to OpenAI

You have two options to connect to OpenAI

1. Use your own account
2. Use our proxy

Password to get a key:

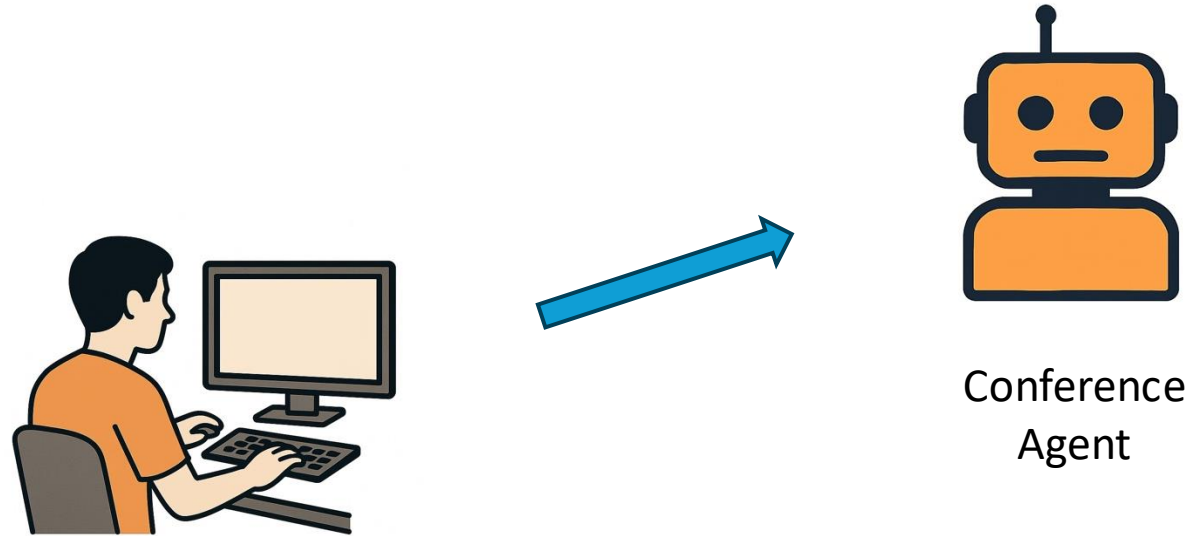
[https://github.com/RAG4J/
jfall-2025-assignments](https://github.com/RAG4J/jfall-2025-assignments)

Plain Java Agent

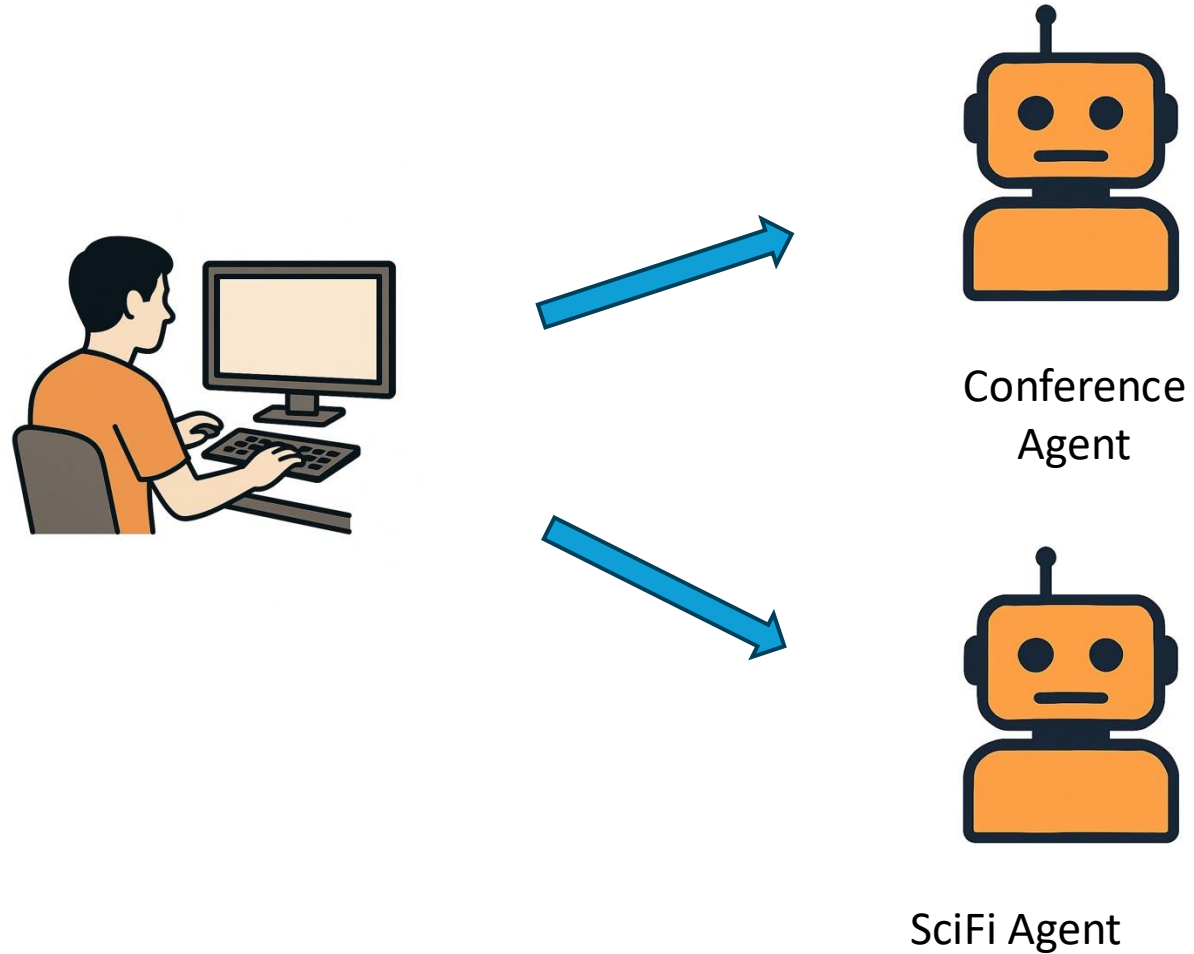
TIME TO WORK ON ASSIGNMENTS



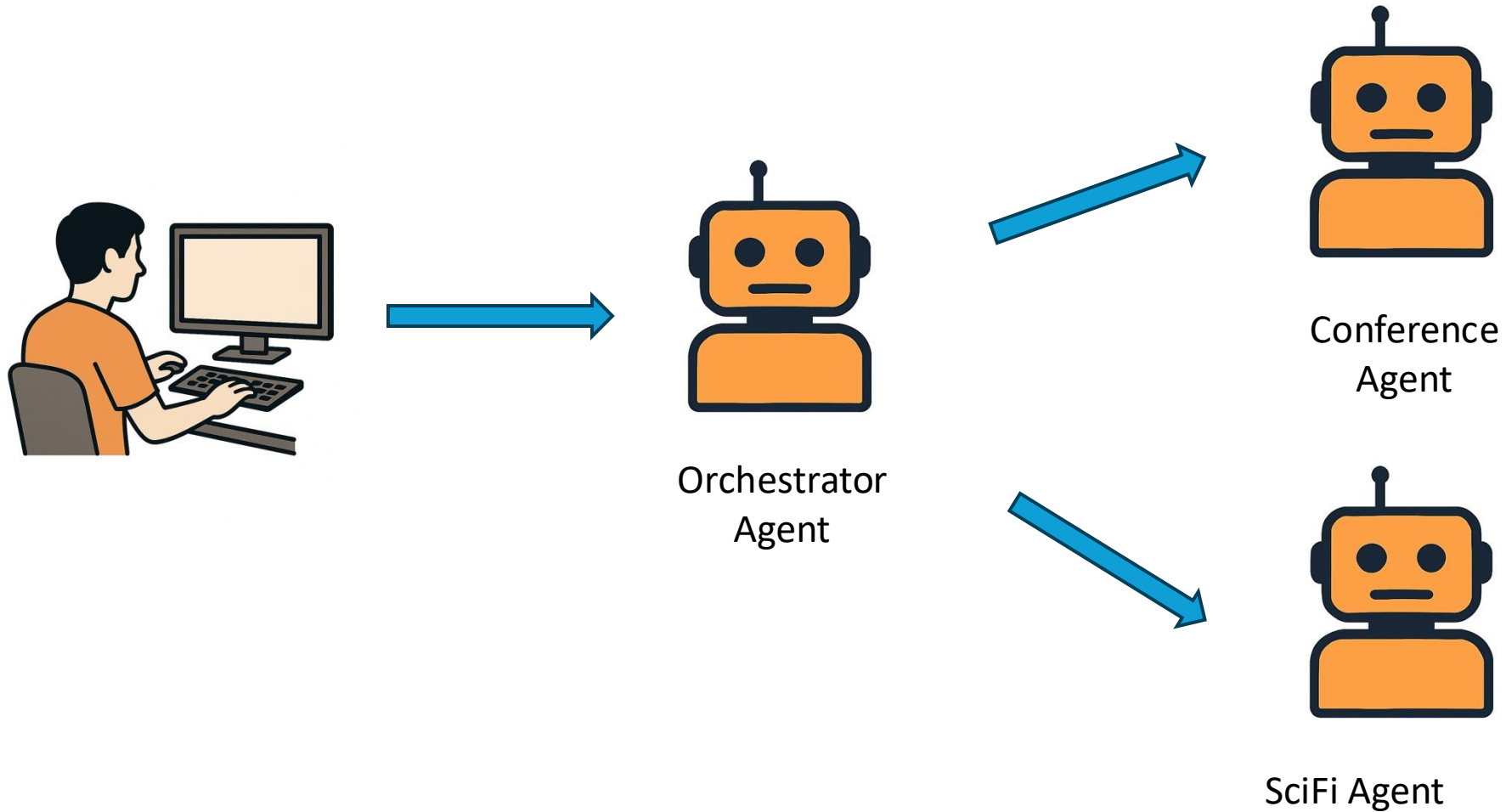
How do Agents work together?



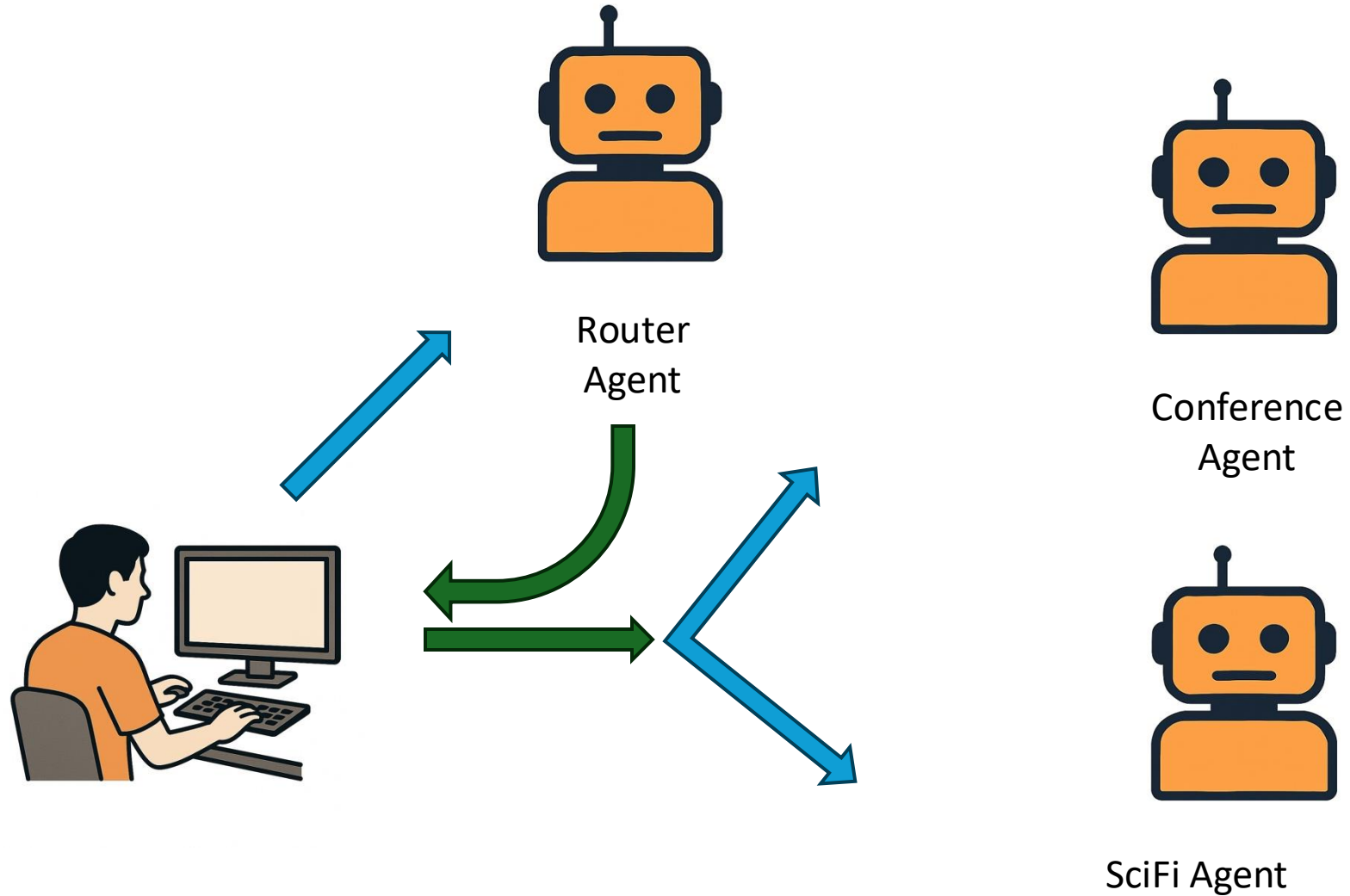
How do Agents work together?



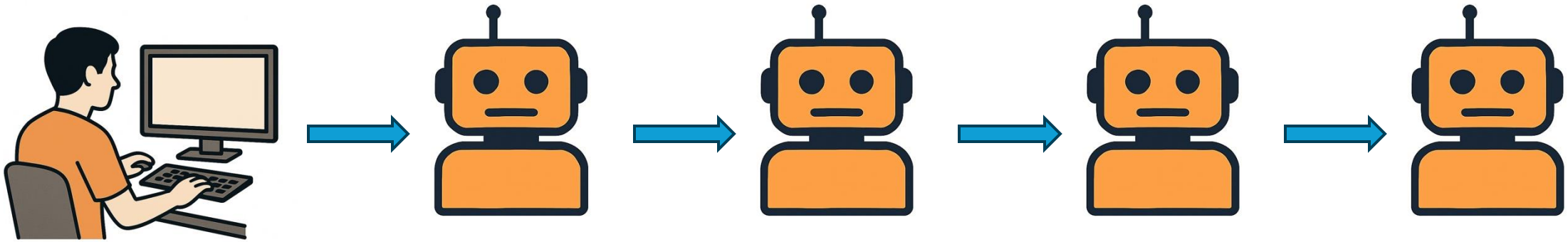
Orchestrator pattern



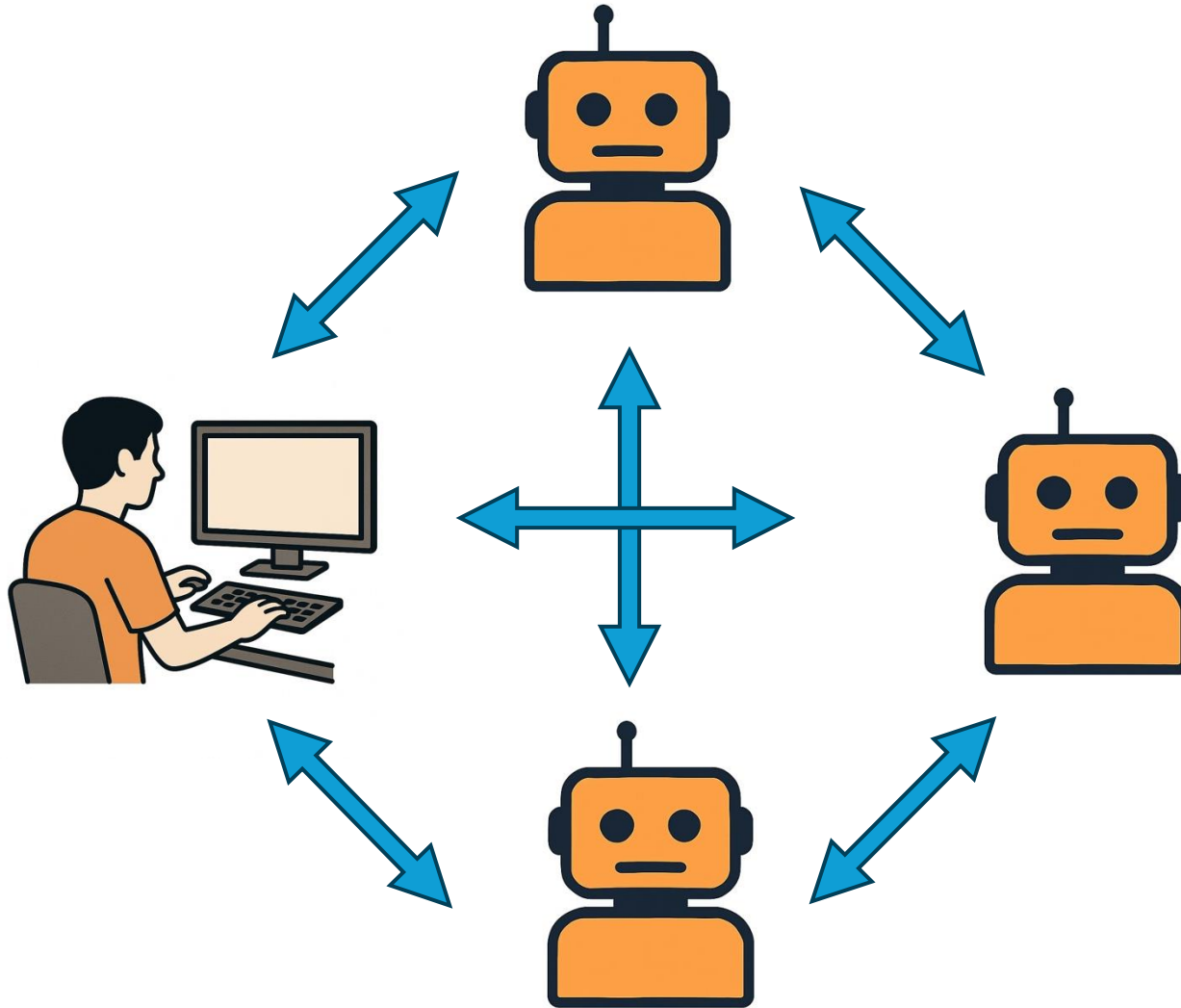
Router pattern



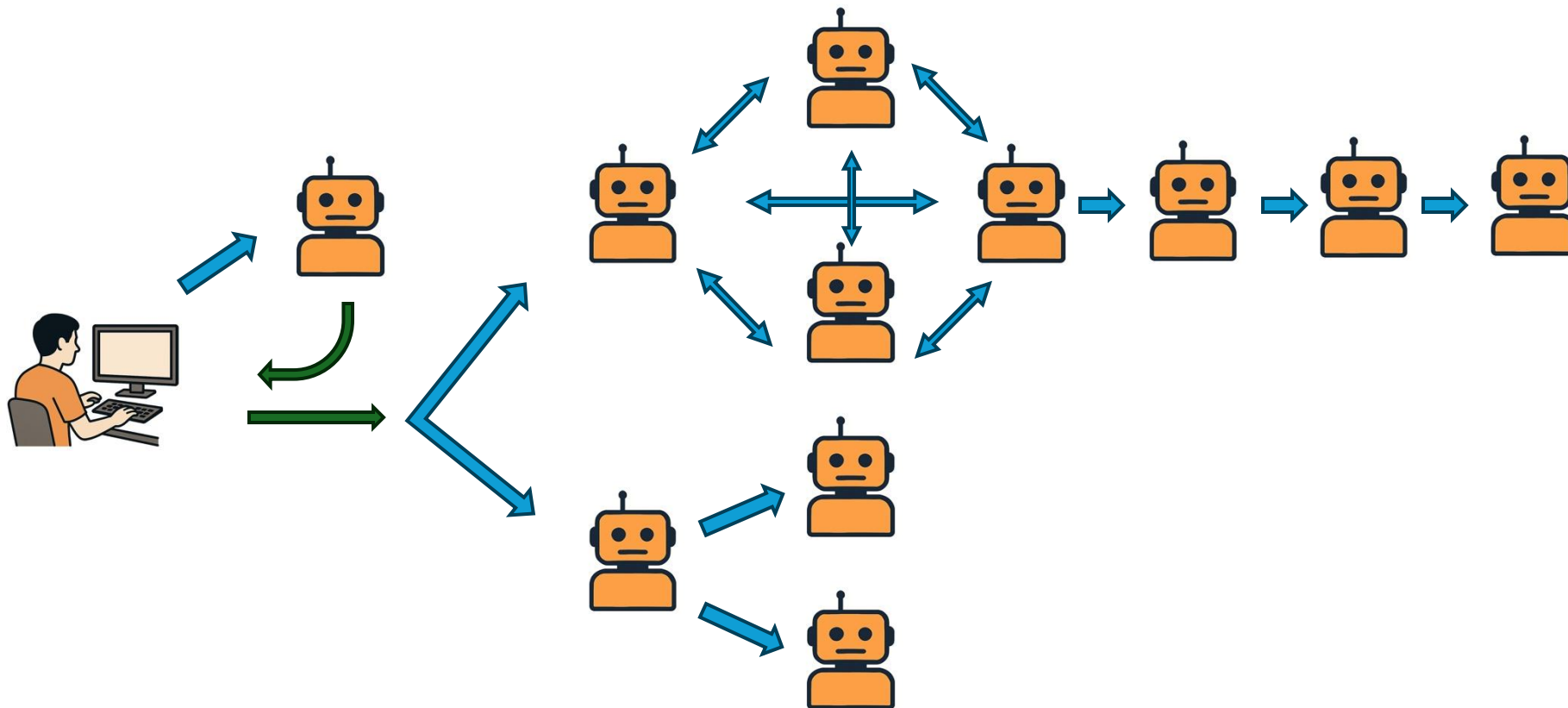
Workflow pattern



Group chat pattern



Chaos pattern?



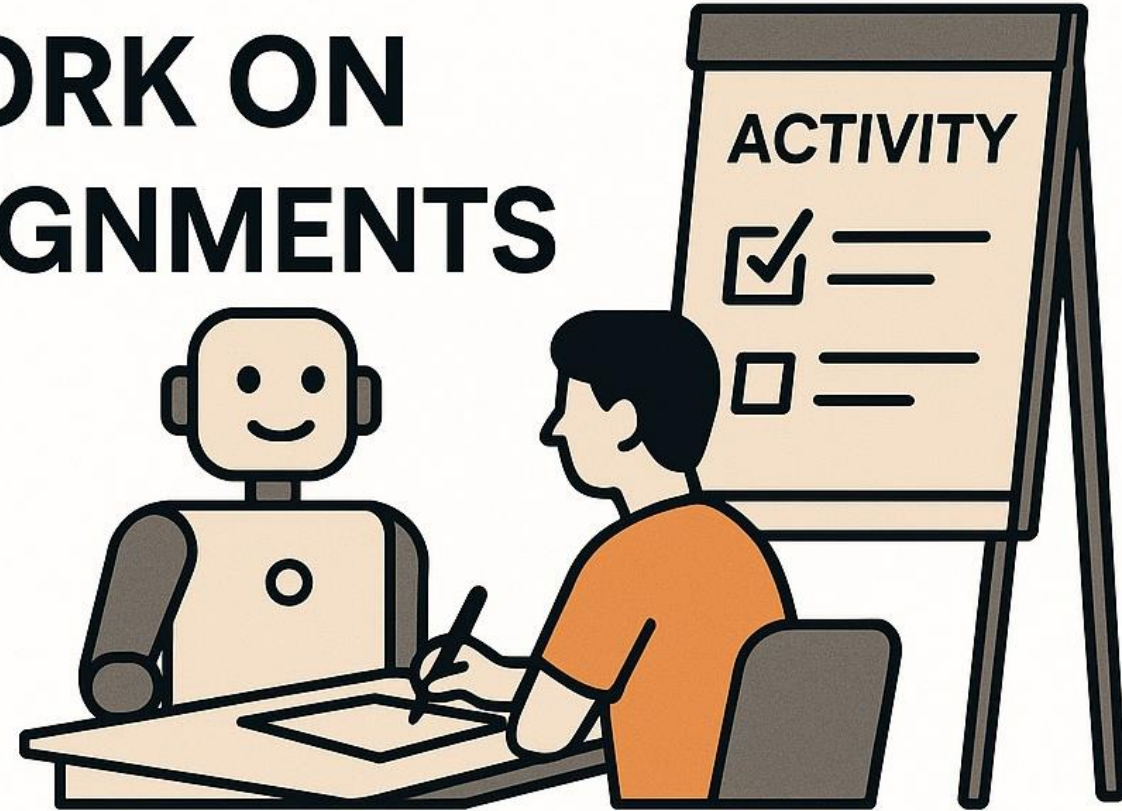
Introduction of Spring AI

- Abstractions and sane defaults
- Works with multiple LLM providers for:
 - Chat, Embeddings, Text to Image, Text to Speech
- Works with a lot of Vector stores
- Has Observability out of the box
- Execution environment for your tools
- Integration with MCP

<https://docs.spring.io/spring-ai/reference/index.html>

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Spring AI



Guardrails

Guardrails are important for agents because they ensure safety, reliability, and alignment with human goals by preventing unintended, harmful, or biased behaviours.

Agentic AI Risk Management

OWASP and IBM Whitepapers

[Agentic AI - OWASP Lists Threats and Mitigations](#)

[IBM Whitepaper: Accountability and Risk Matter in Agentic AI](#)

- Intent Breaking & Goal Manipulation.
- Tool Misuse.
- Memory Poisoning.
- Identity Spoofing & Impersonation.
-

Guardrail Policies

- Policy-first.
- Start simple.
- Make policies readable (non-technical).
- Use layered defence.
- Balance security with user experience.
- Fail closed, not open.
- Align guardrails with business and brand policies.
- Incorporate human oversight strategically.
- Log, monitor and evaluate.

Guardrail Types

- Input filters.
- Output validation.
- Agent and/or Tool restriction.
- Human-in-the-Loop.

Evaluations

The Hidden Key to Reliable Agentic AI

Evaluation isn't the final step; it's the engine of iteration.

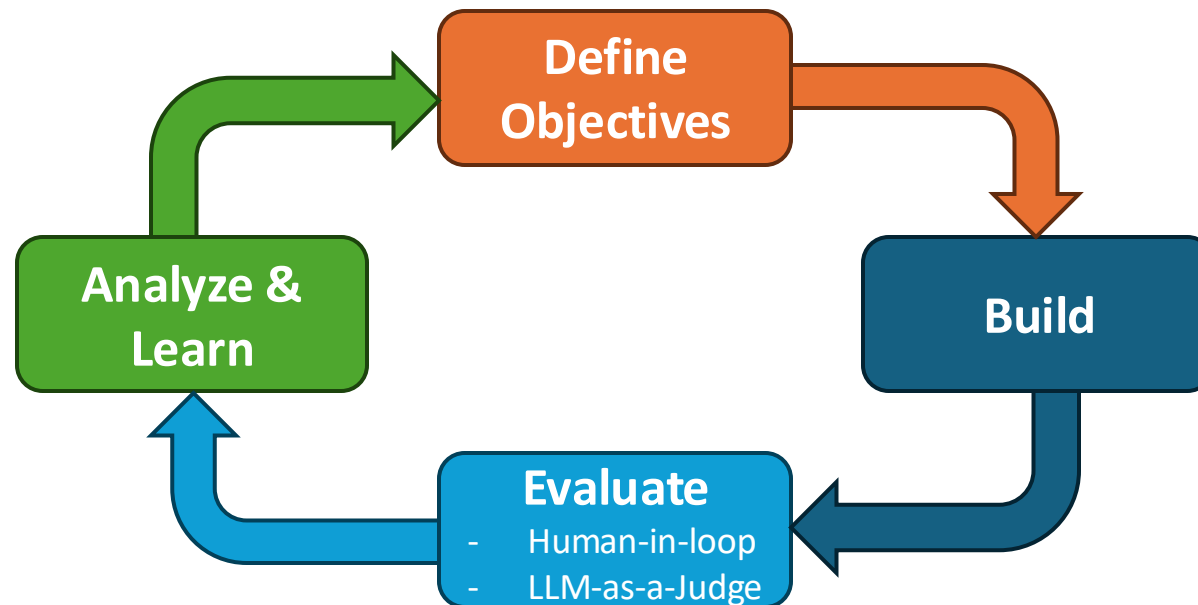
Why Evaluations?

- LLMs are unpredictable, nondeterministic, and subtle in failure.
- No evals = trial-and-error, user complaints, unreliable systems.
- Evals = confidence in changes, systematic improvement.

Core challenges of Evals

- **Data understanding**
→ real users behave unpredictably.
- **Specification gap**
→ hard to encode “good” answers.
- **Inconsistent behaviour**
→ tiny changes, very different outputs.

EDD (Evaluation Driven Development)



Principles for Success

- Start small, keep humans in the loop.
- Look at lots of data, automate only after.
- Build domain-specific evals.

localhost:8081/home

ABP

D

Finish update

Rag4j ~ Agent Evals [springai-multi]

Home Evaluations Runs New Run



Evals Manager

Comprehensive evaluation management for AI systems with human-in-the-loop scoring



View Evaluations

Browse and manage evaluation records with detailed input/output analysis, LLM scores, and human scoring capabilities.

[→ View Records](#)



Manage Runs

Create, configure, and monitor evaluation runs with customizable settings for different AI models and evaluation criteria.

[≡ View Runs](#)

Key Features



LLM Scoring

Automated evaluation with AI-generated scores and detailed reasoning



Human Validation

Human-in-the-loop scoring with detailed reasoning and validation



Detailed Views

Interactive modals and detailed views for comprehensive analysis

Evaluation Records

awesome evaluations

+ New Run

View Run

All Runs

Filter by Run:

awesome evaluations

Showing 5 records from run "awesome evaluations"

Input	Response	LLM Score	LLM Reason	Human Score	Human Reason	Timestamp	Actions
Who is the speaker for a talk... Details	The speaker for the talk abo... Details	Good	The response provides a cle... Details	Good Score	Correct Details	2025-11-06 12:14	
What does Jettro speak abo... Details	Jettro speaks about "From S... Details	Good	The response accurately de... Details	Good Score	Correct Details	2025-11-06 12:14	
How many talks are about AI... Details	There is at least one talk sp... Details	Good	The response correctly iden... Details	Bad Score	It just returned one, but a nu... Details	2025-11-06 12:14	
What talks are about develo... Details	I could not find any conferen... Details	Bad	The response is unhelpful a... Details	Bad Score	There is an IntelliJ talk Details	2025-11-06 12:14	
Who talks about a coding ch... Details	There doesn't appear to be ... Details	Good	The response accurately ind... Details	Bad Score	There is a talk about coding ... Details	2025-11-06 12:14	

Accuracy Metrics

Human Score Accuracy:

40,0%

Percentage of responses rated as "Good" by human evaluators

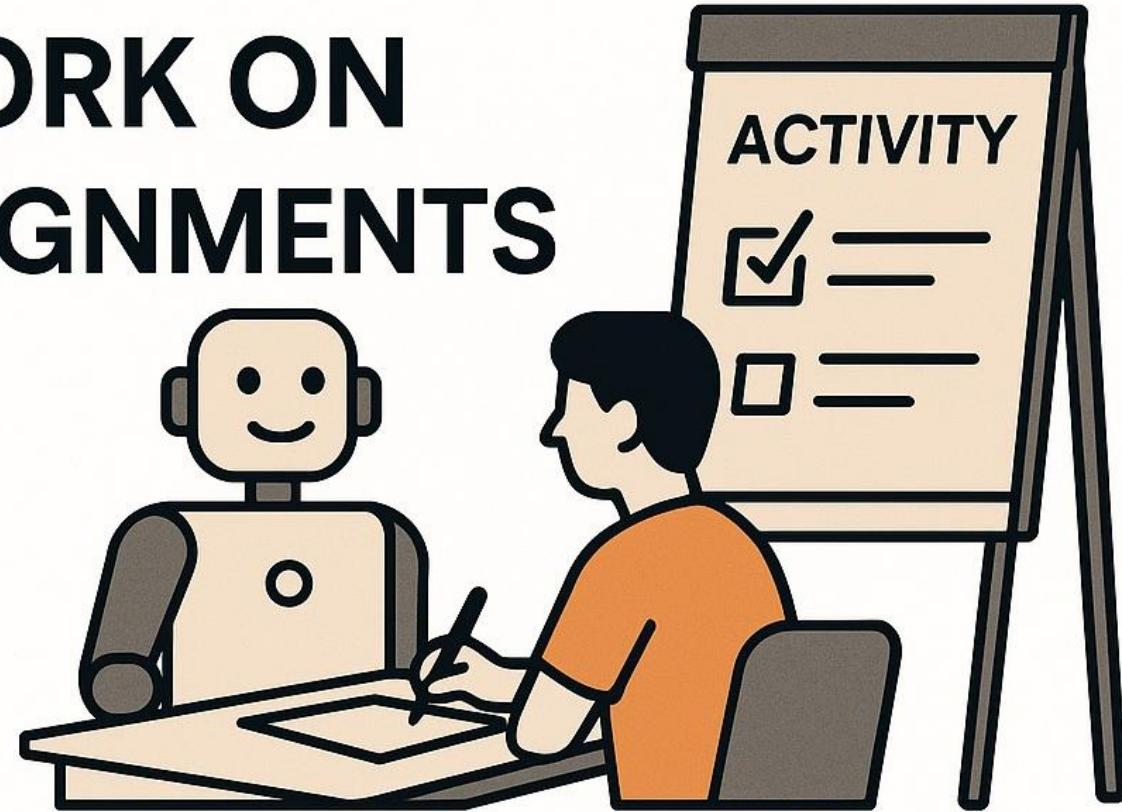
LLM Rating Accuracy:

60,0%

Percentage of cases where human and LLM scores agree

TIME TO WORK ON ASSIGNMENTS

Evaluations and
Guardrails



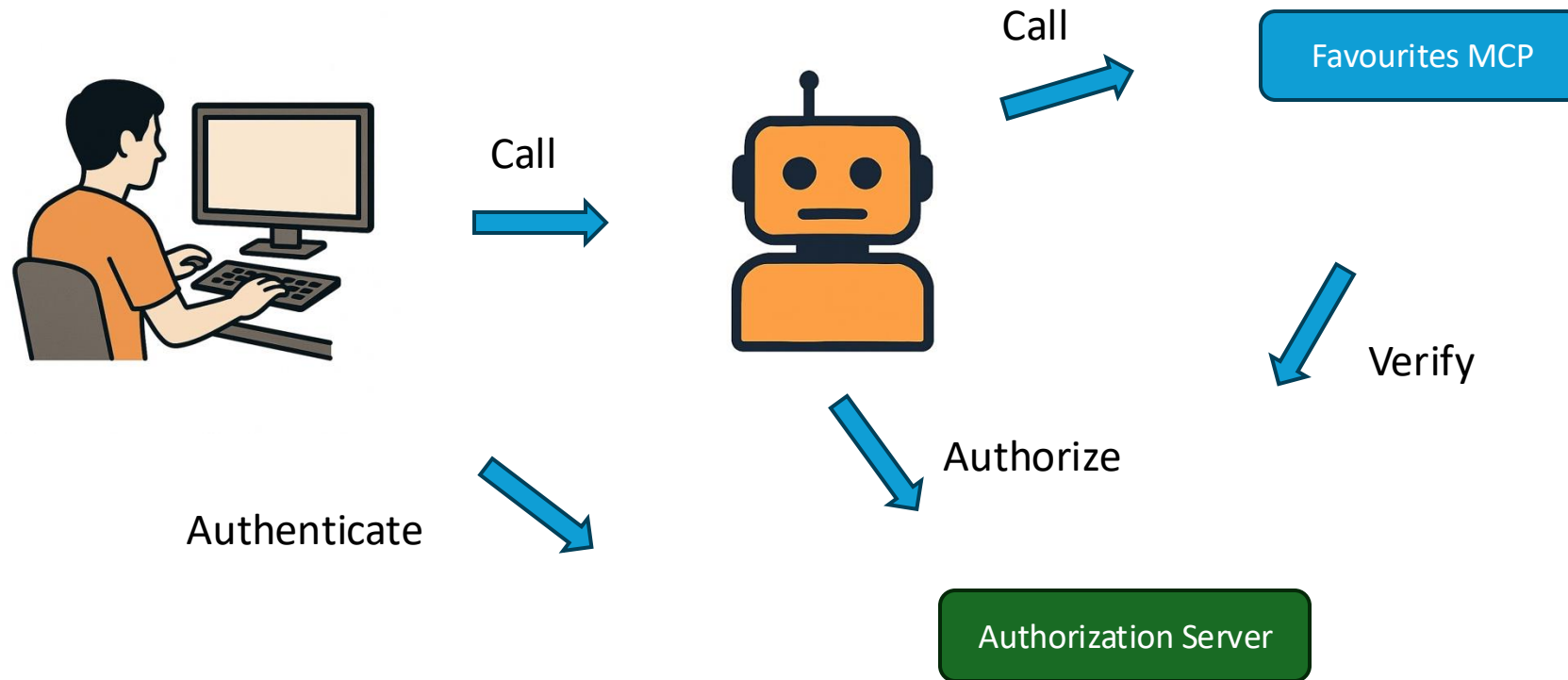
Embabel

High-level introduction

- Builds on top of Spring AI
- GOAP – Goal-Oriented Action Planning.
- Agent – Action – Goal
- Logs are essential for development
- It is new, a lot of developments

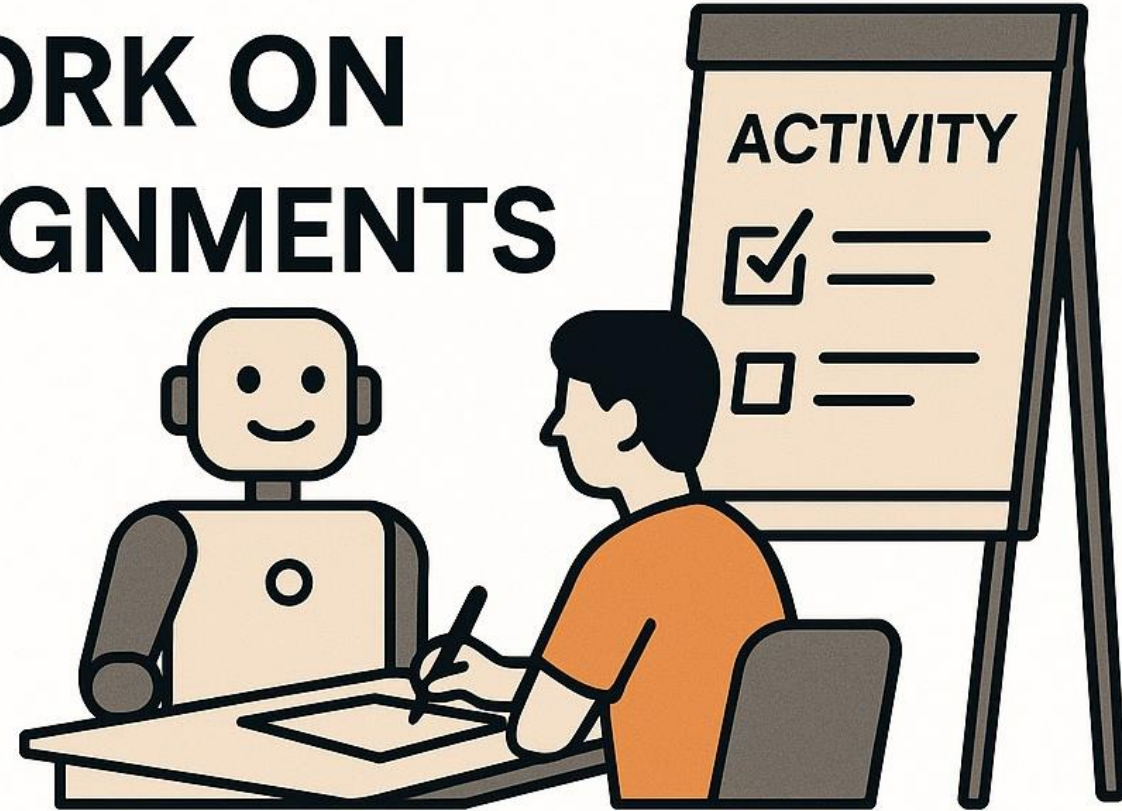


OAuth2 Integration



TIME TO WORK ON ASSIGNMENTS

Embabel + OAuth

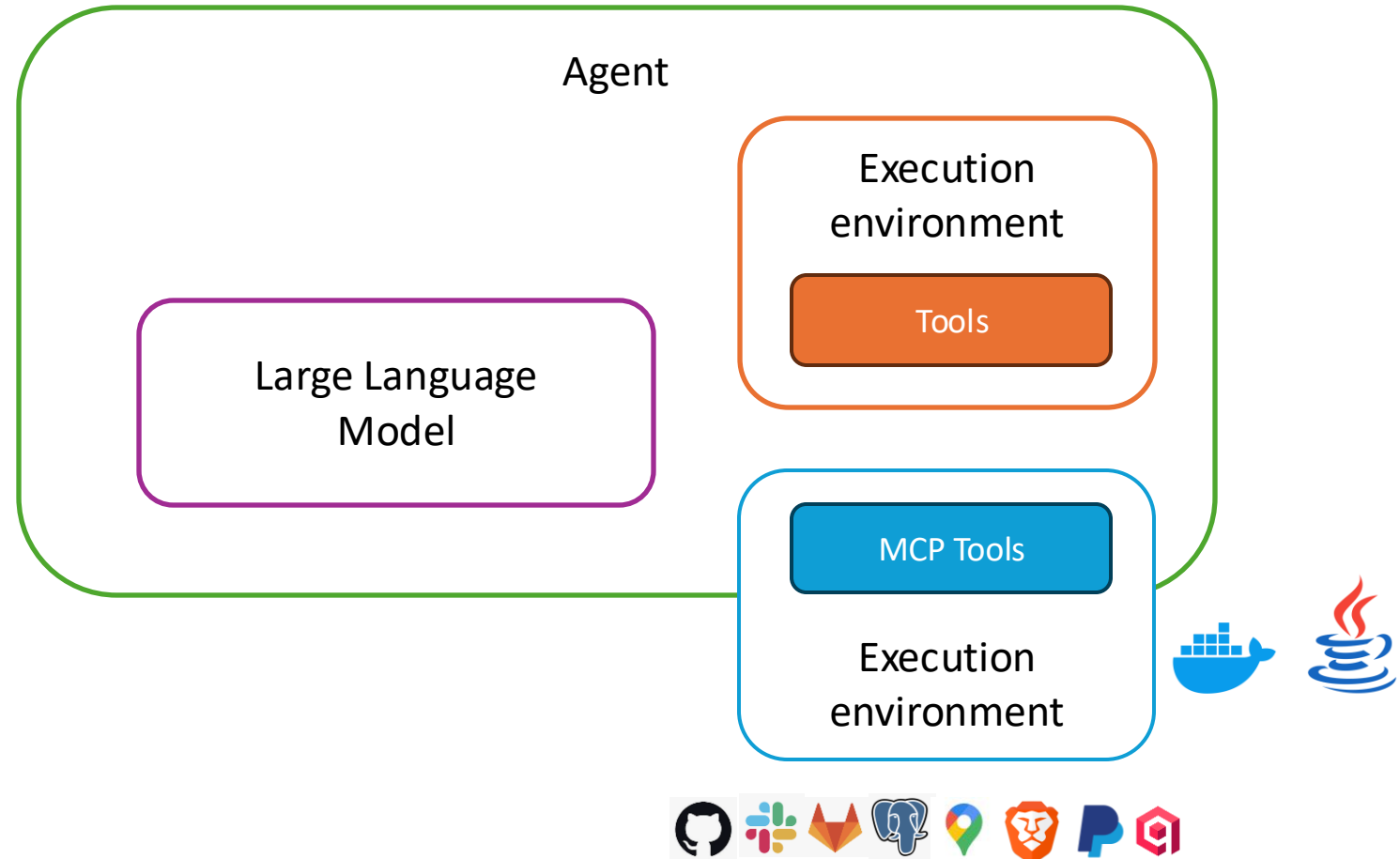


Model Context Protocol [MCP]

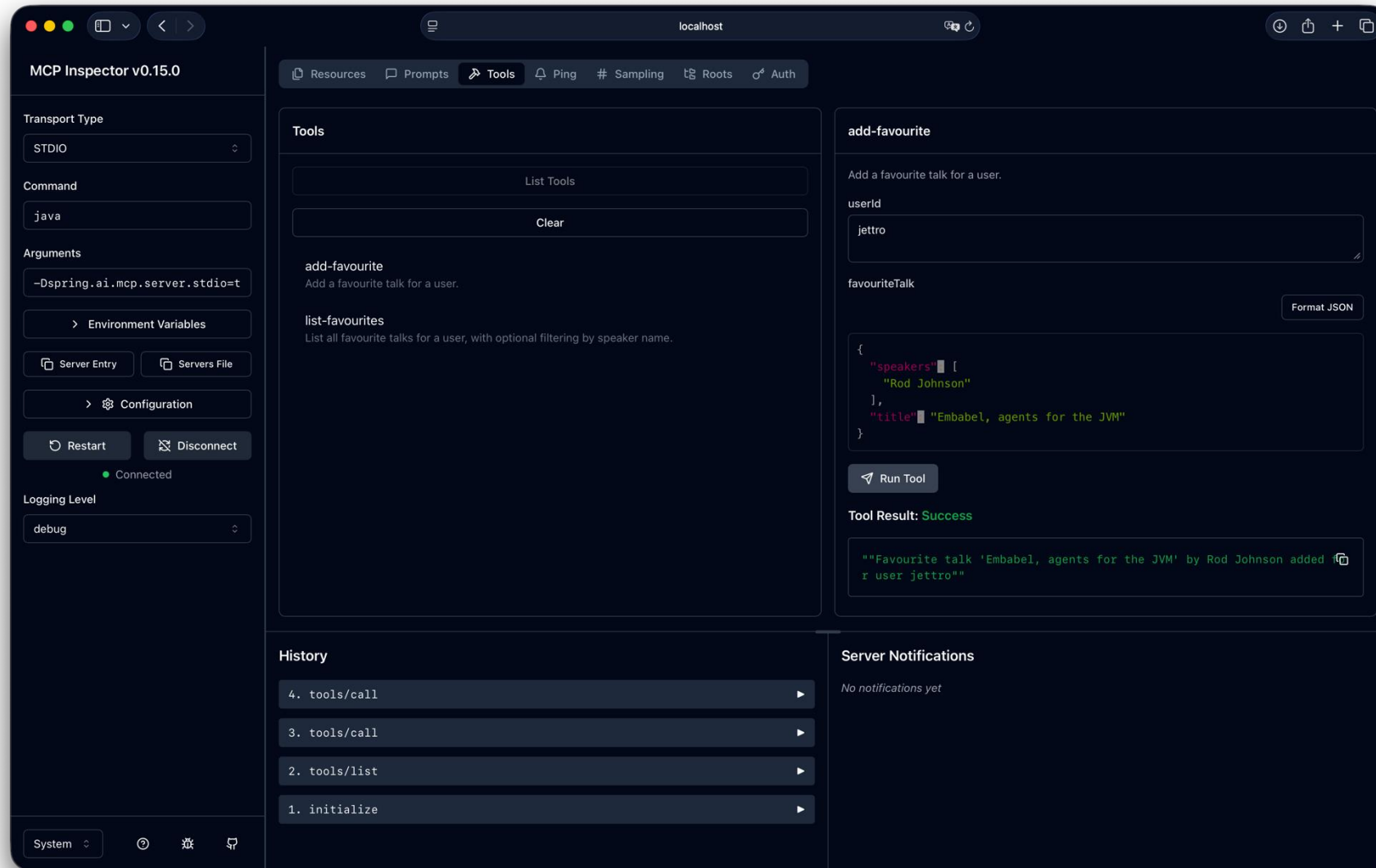
MCP enables you to build agents and complex workflows on top of LLMs and connects your models with the world.

~ <https://modelcontextprotocol.io>

Why do we need MCP?



Use the MCP Inspector



How do you use MCP?

Local access [StdIO]: Single user

```
spring:
  ai:
    mcp:
      client:
        enabled: true
        type: SYNC
      stdio:
        connections:
          favourites:
            command: java
            args:
              - -jar
              - "${FAVOURITES_MCP_JAR:${user.dir}/favourites-mcp/target/favourites-mcp-0.0.1-SNAPSHOT.jar}"
```

How do you use MCP?

Remote running [SSE]: Multi-user

```
spring:
  ai:
    mcp:
      client:
        enabled: true
        type: SYNC
      sse:
        connections:
          favourites:
            url: http://localhost:8081
            sse-endpoint: /sse
```

Is MCP Secure?

Not by default!

- Connection to the MCP
- Connection from the MCP to remote services
- Poorly programmed MCP servers (Without User consent)
- The MCP server provides tools with too many rights
- Malicious MCP servers that trick the agents

TIME TO WORK ON ASSIGNMENTS

Spring AI +
MCP

